

Install PostgreSQL 15

Install and Configure PostgreSQL 15

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Step 1: Install PostgreSQL 15

```
# Install the repository RPM:
dnf install -y
https://download.postgresql.org/pub/repos/yum/reporepms/EL-8-
x86_64/pgdg-redhat-repo-latest.noarch.rpm

# Disable the built-in PostgreSQL module:
dnf -qy module disable postgresql

# Install PostgreSQL:
dnf install -y postgresql15-server
```

Step 2: Create directories

```
mkdir -p /pgTb/pgsql15/trdda
mkdir -p /pgIx/pgsql15/trdda
mkdir -p /pgBackup/pgsql15/trdda
mkdir -p /pgArch/pgsql15/arch
mkdir -p /pgData/pgsql15/data
mkdir -p /pgWal/pgsql15/wal

chown -R postgres:postgres /pg*
chmod -R 700 /pg*
```

Step 3: Change PGDATA directory

```
As root:

[root@lxtrdpdsgv01 ~]# cat
/usr/lib/systemd/system/postgresql-15.service | grep -i
"Environment=PGDATA"
Environment=PGDATA=/pgData/pgsql15/data
[root@lxtrdpdsgv01 ~]#

# systemctl daemon-reload
```

Step 4: Initialize the database

Run as postgres user:

```
/usr/pgsql-15/bin/initdb -D /pgData/pgsql15/data --
waldir=/pgWal/pgsql15/wal
/usr/pgsql-15/bin/pg_ctl -D /pgData/pgsql15/data -l logfile
start
ps -ef | grep postgres
/usr/pgsql-15/bin/pg_ctl -D /pgData/pgsql15/data stop
```

Step 5: Enable automatic start

```
systemctl stop postgresql-15
systemctl enable postgresql-15
systemctl start postgresql-15
systemctl status postgresql-15
```

Step 6: Enable archive mode and change archive location

As postgres user:

```
cp /pgData/pgsql15/data/postgresql.conf
/pgData/pgsql15/data/postgresql.conf.bkp
grep -v '^#' /pgData/pgsql15/data/postgresql.conf.bkp | grep
'^[A-Za-z0-9]' > /pgData/pgsql15/data/postgresql.conf
```

```
edit -- /pgData/pgsql15/data/postgresql.conf
```

```
# Added by DBA
listen_addresses = '0.0.0.0' # ipv4 only
max_wal_senders = 10
max_replication_slots = 10
wal_level = 'replica' # or 'logical'
hot_standby = on
archive_mode = on
archive_command = 'cp %p /pgArch/pgsql17/arch/%f'
# shared_preload_libraries = 'repmgr'
wal_log_hints = on
# End
```

Step 7: Reload PostgreSQL service

```
systemctl stop postgresql-15.service
systemctl start postgresql-15.service
```

Step 8: Verify configuration

```
sed -i "s/PGDATA=.* /PGDATA=\/pgData\/pgsql15\/data/"
.bash_profile

psql

SHOW data_directory;
SHOW archive_mode;
SHOW archive_command;
SHOW archive_timeout;
ls -ld $(psql -t -c "SHOW data_directory")/pg_wal
SELECT pg_switch_wal();
ls -lrth /pgArch/pgsql15/arch/

SELECT f AS wal_file,
       (pg_stat_file('/pgArch/pgsql15/arch/' || f)).size AS
size_bytes,
       (pg_stat_file('/pgArch/pgsql15/arch/' ||
f)).modification AS modified_time
FROM pg_ls_dir('/pgArch/pgsql15/arch') f
WHERE f ~ '^[0-9A-F]{24}$'
ORDER BY modified_time DESC;
```

Step 9: Add firewall rules

```
# need sudo or root user
firewall-cmd --add-port=5432/tcp --permanent --zone=public
firewall-cmd --reload
firewall-cmd --list-ports
```

Caution: Your use of any information or materials on this website is entirely at your own risk. It is provided for educational purposes only. It has been tested internally, however, we do not guarantee that it will work for you. Ensure that you run it in your test environment before using.

Thank you,

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